

majority of us."

On the contrary, Srivasta Raghunath, technical lead - Analog Systems, Texas Instruments restricts himself to the new standards. He says, "Since A4WP is still evolving, there has been no standardisation and no particular tools developed yet. This is still in R&D stages and will be evolving in the coming years."

Future: Industrial use

Although the main focus of wireless charging has been small portable devices, it cannot be limited to these devices alone. The main aim of this type of charging technology is to transfer maximum power to the receiver in a wire-free environment.

All electric transport buses that have been introduced on a trial basis, on the streets of Milton in UK for five years, use the wireless charging technology on a larger scale. Although the buses are completely charged at night, they need top up charges after every route. The buses receive the top-up charges from the top-up stations that appear as plates on the road.

After completing one route, the bus is parked on that plate where the driver lowers the receiver plate from the bottom of the bus to 4 cm below the road surface. The bus is charged only for ten minutes before starting on the next route. It follows the same magnetic flux technology for wireless charging. Though these buses are currently being used on a trial basis, it is not long before wireless charging will come to the mainstream.

"We envision a world in which wireless charging would accelerate the adoption of clean, green electrified vehicles," says WiTricity CEO Eric Giler in one of the reports. With Toyota deciding to license intellectual property from WiTricity to offer wireless charging on both hybrid-

Strike 100 in a few minutes

The development of high-rated, electro-chemical batteries is making wireless charging more practical and more useful. Its high-rate capability charges the battery completely within few minutes. Stu says, "In fact, even with today's well-known battery chemistries, you can 'tune' the design of the cell to give up a little in energy density in order to support a high charge rate."

Two-in-one

MOCREO shifts the electrical charging contacts from the socket to the back of the device. It is a portable charging pad that weighs only 240 grams and plays a double role in the family of wireless chargers. It has two USB ports that facilitate wired charging of two devices even when it is wirelessly charging the third one. It also acts as a portable power bank and stores about 10000mAh power that can be used to power the devices using the USB plug-in port.

Driving wireless charging technology forward

A4WP is accepting the inductive charging solution pioneered by PMA, leaving behind no working groups to develop another inductive charging standard. While on the other hand, PMA is accepting the resonant charging solution pioneered by A4WP (called Rezenace) as the industry standard for resonant charging. This way even A4WP is no longer going to have working groups that would develop another resonant charging solution. Itay says, "The whole point of this is to harmonise the various recognised industry standards that enable consumers to power up wirelessly." They have developed a common certification to accelerate the use of wireless charging in more devices and have collaborated on their API (application programming interface) for network service management. Qualcomm joined this group two weeks after joining WPC, leading the roles of three different bodies for wireless charging.

A technology can never improve with just one merger or collaboration. There has to be continuous improvements along with partnerships and investments to make the technology widely known and acceptable. Samsung's Electro-Mechanical division recently partnered with PowerbyProxi to develop new charging devices. Samsung, being a member of the A4WP consortium, is looking forward to promote the Qualcomm's WiPower standard based on resonance.

electric and fully-electric vehicles, Eric adds, "To have Toyota, the world's leading carmaker, licensing our intellectual property, underscores the importance of the technology."

In future we will not only see the extensive use of wireless charging technology but also see some enhancements in the standards enabling this technology. Although Qi is typically an inductive charging technology, WPC is developing a resonance extension to its specifications. Resonant extension to Qi is expected to be backward compatible with more than 40 million products and devices globally, including the 60 models of Qi-compatible devices. In one of

the reports, John Perzow, vice president of Market Development for the Wireless Power Consortium said that, Qi is constantly enhancing to give away the best wireless charging experience.

Stu Lipoff, IEEE fellow, said in a conference, "One vision is to have NFC RFID in the phone that provides the local power company with billing information so that they can charge your credit card for power purchased as you walk by the public outdoor wireless charging infrastructure." He added, "With the development of extended distance wireless charging, it even becomes possible to charge the devices by walking outside on the street in proximity to public portable power infrastructure."

Itay says, "PMA is already working to globalise this wireless charging by integrating more technologies at physical layer. We are working on to provide networking standards and to deploy public hotspots for wireless charging." ●

The author is a technical journalist at EFY

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The cost of the product is mainly due to the testing and compliance efforts that go into the product. The semiconductor content does not increase price of the product but the R&D that goes into developing the product increases the price.” —Srivasta Raghunath, TI